Supersymmetry changes this drastically because bosons—particles whose spin is a whole number (named after the Indian physicist Satyendra Bose)—and fermions—particles whose spin is half of a whole (odd) number (named after the Italian physicist Enrico Fermi)—tend to give cancelling quantum-mechanical contributions. Like opposite ends of a seesaw, when the quantum jitters of a boson are positive, those of a fermion tend to be negative, and vice versa. Since supersymmetry ensures that bosons and fermions occur in pairs, substantial cancellations occur from the outset—cancellations that significantly calm some of the frenzied quantum effects. It turns out that the consistency of the supersymmetric standard model—the standard model augmented by all of the superpartner particles—no longer relies upon the uncomfortably delicate numerical adjustments of the ordinary standard model. Although this is a highly technical issue, many particle physicists find that this realization makes supersymmetry very attractive.

The third piece of circumstantial evidence for supersymmetry comes from the notion of grand unification. One of the puzzling features of nature's four forces is the huge range in their intrinsic strengths. The electromagnetic force has less than 1 percent of the strength of the strong force, the weak force is some thousand times feebler than that, and the gravitational force is some hundred million billion billion billion (10-35) times weaker still. Following the pathbreaking and ultimately Nobel Prize-winning work of Glashow, Salam, and Weinberg that established a deep connection between the electromagnetic and weak forces (discussed in Chapter 5), in 1974 Glashow, together with his Harvard colleague Howard Georgi, suggested that an analogous connection might be forged with the strong force. Their work, which proposed a "grand unification" of three of the four forces, differed in one essential way from that of the electroweak theory: Whereas the electromagnetic and weak forces crystallized out of a more symmetric union when the temperature of the universe dropped to about a million billion degrees above absolute zero (1015 Kelvin), Georgi and Glashow showed that the union with the strong force would have been apparent only at a temperature some ten trillion times higher—around ten billion billion billion degrees above absolute zero (1028 Kelvin). From the point of view of energy,